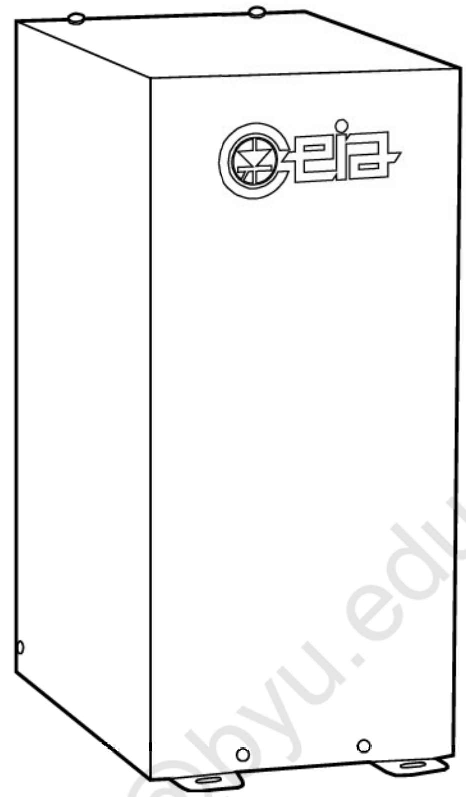


Inductive Heaters

PowerCube 90 and 180

Installation, Use and Maintenance Manual

Document	Date	Software	Hardware
FI040K0002v1100hUK	18/03/2021	4.9xx	1.xx

Read this manual carefully before installing, operating or carrying out maintenance on the device. Keep the manual in a safe place for future reference, and in perfect condition. This manual must accompany the device described therein in the case of change of ownership, and until the device is broken up.

SYMBOLS

The equipment is marked with this symbol wherever the user should refer to this manual in order to avoid possible damage. The same symbol appears in the manual at points where warnings or particularly important instructions, essential for safe, correct operation of the device, are given.

The equipment is marked with this symbol in the areas where there is dangerous voltage. Only trained maintenance personnel should carry out work in these areas. The same symbol appears in the manual at points where warnings essential for safe concerning these areas.

The equipment is marked with this symbol in the areas where there could be present very high temperatures. The same symbol appears in the manual at points where warnings essential for safe, are given.

INTENSE ELECTROMAGNETIC FIELDS may have an effect on pacemakers. Persons wearing pacemakers, cardiac defibrillators and other support devices must not operate close to the inductor or the heating inductors of the machine. These individuals must consult their personal doctor if they are to operate other than as stated above.

This symbol appears in the manual at points where suggestions, additional information or other relevant notes are given.

REVISION TABLE

Rev.	Date	Author	Reference	Description
1000	25/06/2008	TP2 - SP	-	First emission
1010	15/07/2009	TP2 – SP	6.1	CE certificate
1020	28/10/2014	TP2 – SP	-	Small corrections
1030	13/12/2017	TP2 – SP	6.1	CE certificates
1040	27/04/2020	DTP – SP	2.3.3	Small correction
1100	18/03/2021	DTP – SP	-	Software update

Warranty terms

The warranty on all C.E.I.A. products, extended to the period agreed with the Sales Department, is applicable to goods supplied from our factory, and for every constituent part thereof, with the exception of the batteries. Any form of tampering with the device, and in particular opening its container, is strictly forbidden and will invalidate the warranty.

Customer Satisfaction Report

Your suggestions and comments on the products and services offered by CEIA and its distribution network are extremely important for improving our procedures.
We would ask you to send them to us by compiling and returning the form available at:

http://www.ceia-power.com/satisfaction

Thank you for your kind interest and co-operation.

CONTENTS

Read this manual carefully before installing, operating or carrying out maintenance on the device.

1 SAFETY INSTRUCTIONS - WARNINGS

1.1 Warnings

1.1.1 General

- Heating processes which have been developed over the years may be employed with a high degree of safety provided that all the safety regulations suggested by common sense and by the manufacturer's recommendations are observed. It is therefore essential that heads of staff safety make sure that all operators read this manual before being allowed to use the device.
- Follow the instructions contained in this manual for all operations relating to installation, use and maintenance of the device. CEIA cannot be held responsible for any damage resulting from procedures which are not expressly indicated in this manual, or from any lack of attention, either partial or total, of the procedures described therein.
- Whenever there is any suggestion that the level of protection has been reduced, the device should be taken out of service and secured against any possibility of unintentional use, and authorised service technicians should be called.
  The level of protection is considered to have been reduced when:
  - *the device shows visible signs of wear and tear, especially on parts that ensures the device protections (boxes, gaskets, glands, fixing screws, etc.);*
  - *the electrical cables or the equipotential ground connections are damaged.*
  - *the ground connections are not compliant;*
  - *the device has suffered mechanical or electrical stress (shocks, bumps, etc..);*
  - *the device does not operate correctly, respect on how described in this manual;*
  - *the device has been stored for long periods in sub-optimal conditions;*
  - *the device has suffered severe stress during transport;*
  - *the inside of the device has come into contact with corrosive substances (water infiltration, etc.).*
  - *the device was present inside an environment where a fire or an explosion took place.*
- This equipment complies with European (CE) and International Standards on Electrical Safety and Electromagnetic Compatibility. However, CEIA will not accept any liability for damage to objects or people due to improper use or to specific incompatibility with equipment of a different nature.

1.1.2 Installation

The device is designed to be inserted as a component part of a heating system; the safety devices needed to protect the operators must therefore be designed and fitted by the installer according to the type of application in question.

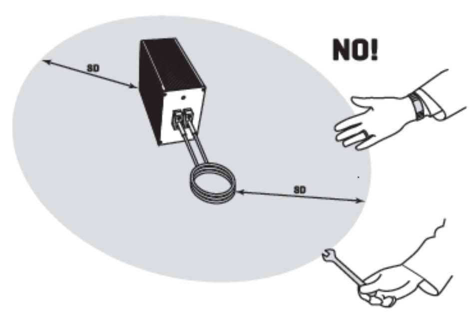
- Before the equipment can be used it must be fitted with guards and protections to prevent direct contact and improper use.
- Ensure that there is adequate fire-fighting equipment sufficiently close to the working area.
- Choose the installation site carefully. Avoid placing the device in locations where it may be directly exposed to sunlight or in places that are close to sources of heat. In addition, avoid places that are subject to vibrations, dust, humidity, rain and excessively high or low temperatures.
- Handle the device with care and without excessive force during installation, use and maintenance.
- The operator must be able to access the controls of the device easily, without running personal risks.
- To avoid damage due to lightning, fit the power supply line with appropriate surge suppressors.

- This device contains electrical and electronic components, and may therefore be susceptible to fire. Do not install in explosive atmosphere or in contact with inflammable material.
- After installation the device should be stable, and not subject to vibration or accidental movement. All connecting cables should be properly fastened down, in order to avoid knocks and accidental damage and to obtain optimum performance.
- If the device is to be powered via an external autotransformer to regulate the voltage, ensure that the common terminal of the autotransformer is connected to the neutral of the power-supply circuit.
- The electrical plug must only be inserted into a socket containing a grounding contact. Any interruption in the protective conductor either inside or outside the device, or the detaching of the protective grounding terminal, renders the device dangerous. Intentional interruption is forbidden.
- Before providing the device with electrical power, check that the voltage from the power mains corresponds to the voltage shown on the device's electrical specifications plaque.
- Only connect the device to the power mains after having made all the connections needed for its complete installation.

1.1.3 Use

INTENSE ELECTROMAGNETIC FIELDS may have an effect on pacemakers. Persons wearing pacemakers, cardiac defibrillators and other support devices must not operate close to the inductor or the heating inductors of the machine. These individuals must consult their personal doctor if they are to operate other than as stated above.

- Do not braze work pieces which have been in contact with toxic, combustible materials. You should thoroughly clean these work pieces before proceeding with brazing.
- Maximum care must be taken to avoid incandescent metals coming into contact with combustible materials.
- Ensure that there is adequate fire-fighting equipment sufficiently close to the working area.
- All combustible materials must be removed from the working area and placed at a safe distance.
- During heating phase, maintain a minimum safe distance from operator body parts and the activated heating heads and relative inductors, in order to avoid any risk of burning due to the heating, even without direct contact, of metal objects in contact with the operator (rings, tools, watches, etc…) and not to exceed the limits set by current regulations on the maximum level of human exposure to electromagnetic fields.



The **SD** safety distance, depending on the type of inductor used, can reach up to **80 cm**.

The shape of the inductor in the figure is an example and is provided as a guide only.

- When the generator is operating, the heating inductor is powered up and carries hazardous high voltages (as defined by standard CEI EN 61010-1, section 6,3,1). Avoid all direct or indirect contact with metal objects.
- During normal use the heater does not product sparks. However, electrical discharge can take place close to the heating inductor during heating in the event of accidental contact between the inductor and any metal parts connected to ground (earth).
- During normal use, the heater does not produce sparks. Electrical discharges may however occur during soldering, near the heating inductor, in the event of accidental contact between the inductor and metal parts with a ground connection.
- After soldering, ensure that the piece that has been worked upon is allowed sufficient time to cool before handling it or bringing it into contact with combustible material.

- All electrical shocks are potentially fatal. Never touch parts charged with mains power. Keep clothes and body dry and never work in wet environments. If you ever have cause to believe that the device or part of it has accidentally become electrically charged, stop heating operations immediately and do not use the device until the problem has been located and resolved by qualified personnel.
- Always remove the plug by hand when disconnecting the power supply cable, never by pulling on the cable.
- This device contains electrical and electronic components. Do not use water or foam in the case of fire when the device is powered up.

1.1.4 Maintenance

- Do not wash the device with liquid detergents or chemical substances. Any cleaning should be done using a slightly-damp, non-abrasive cloth.
- Before any maintenance, cleaning or movement, the device must first be disconnected from all sources of power.
- Read the chapter on “Maintenance” carefully before calling the service centre. Whatever the problem, only specialised service personnel authorised to work with CEIA equipment should be called.
- Any damaged parts of the device should be replaced with original components only.
- Any maintenance or repair of the device while open and energised should be avoided, and in any case should only be carried out by trained personnel who are fully aware of the risks which the operation entails, following the instructions given in the Maintenance section.

1.2 Residual risks

1.2.1 Use

The equipment has been designed to operate as a component of a heating system and/or inside a machine. It is the installer's responsibility to design and install the safety devices necessary to protect operator health and safety.

- When the generator is operating, the heating inductor is powered up and carries hazardous high voltages (as defined by standard CEI EN 61010-1, section 6.3.1). Avoid all direct or indirect contact with metal objects.
- When the generator is operating, safety devices must prevent direct and indirect contact with the heating inductor and the heating head. Take precautions to prevent any direct or indirect contact with the heating inductor and the heating head. Implement measures to maintain a minimum safe distance up to **80 cm for any metal object in contact with the operator and parts of the body.**
  Below these distances, even when there is no direct contact, the limits set by the Standards for Human Exposure to electromagnetic fields may be exceeded.

INTENSE MAGNETIC FIELDS. NO PERSONS WITH LIFE SUPPORT DEVICES. Persons wearing pacemakers, cardiac defibrillators and other life support devices must not operate close to the inductor or the heating inductors of the machine. Persons wearing pacemakers who may need to approach this equipment should consult their medical practitioner beforehand.

1.2.2 Maintenance

- The capacitors within the device may remain charged even after the device has been disconnected from all sources of power. Wait 60 seconds before beginning repair work on the device.

1.3 Correct and Forbidden use of the device

1.3.1 Correct use of the device

The device is designed to be inserted as a component part of a heating system; the safety devices needed to protect the operators must therefore be designed and fitted by the installer according to the type of application in question.

- The PowerCube 90, and 180, is an induction heater designed and constructed for heating metal work pieces. All other uses are strictly prohibited.

1.3.2 Forbidden use of the device

- During heating phase, maintain a minimum safe distance from operator body parts and the activated heating heads and relative inductors, in order to avoid any risk of burning due to the heating, even without direct contact, of metal objects in contact with the operator (rings, tools, watches, etc…) and not to exceed the limits set by current regulations on the maximum level of human exposure to electromagnetic fields.
  The safety distance, depending on the type of inductor used, can reach up to **80 cm**.
- The heating phase shall not be started, for any reason, in presence or in the nearby of tanks under pressure, explosive or inflammable atmosphere or liquids.
- Any different use, from the specifications of paragraph *1.3.1 – Correct use of the device*, is forbidden.

2 DESCRIPTION

2.1 General description

The **PowerCube** is a series of generators dedicated to the soldering, without contact, of metal components.
The core of the device consists of an advanced solid state generator, entirely controlled by microcomputer. This provides power to two miniaturised inductive heads which are easily positioned even in restricted space.
An original energy-transfer solution (patented) allows a rapid heating of the metallic details with a minimum absorption of power from the mains.
The **PowerCube** generators conform to the international standards currently applicable for electrical safety, E.M.C. and to the applicable CE regulations.

2.1.1 Description

The basic configuration of the Inductive Heaters **PowerCube** is as follows:

- a heating head;
- a microcomputer-controlled generator.

The device can be connected to a controller of the **Power Cube Family**.

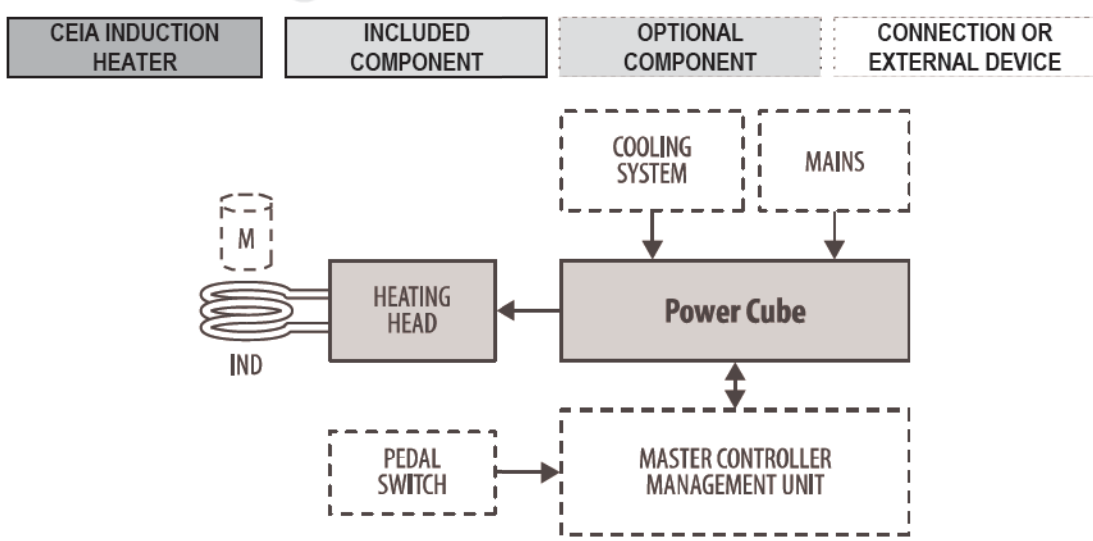
The Generator, suitably configured, can be used in the following configurations:

- Single unit, for power outputs up to 12 kW,
- 2 units connected in parallel and synchronised, to obtain a power output of 24 kW,
- 4 units connected in parallel and synchronised, to obtain a power output of 48 kW.

2.2 Functional Diagram

An inductor with a suitable shape (**IND**), connected to the head emits an electromagnetic field. Metals located within the inductor heat up due to the currents induced by this magnetic field:

> The shape of the inductor depends on the type of application; CEIA does not physically supply the head, which must be produced by the heating system constructor.



*Block diagram of a heating workstation **PowerCube 90** or **180**.*
***M**: metal object to be heated; **IND**: heating inductor*

2.3 Components

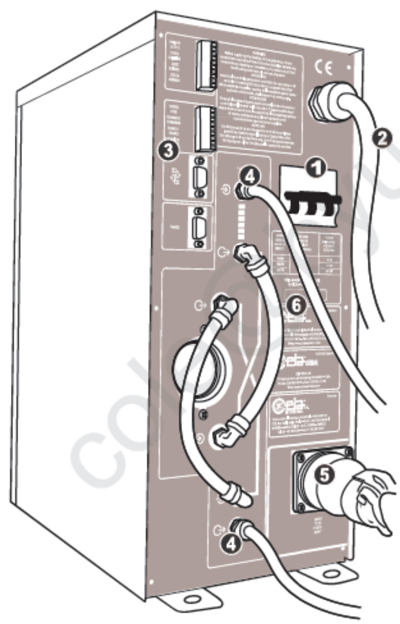
2.3.1 PowerCube 90 and 180

The **PowerCube 90** and **180** generators consists of a central electronic unit which includes a power supply stage and a medium frequency generator.

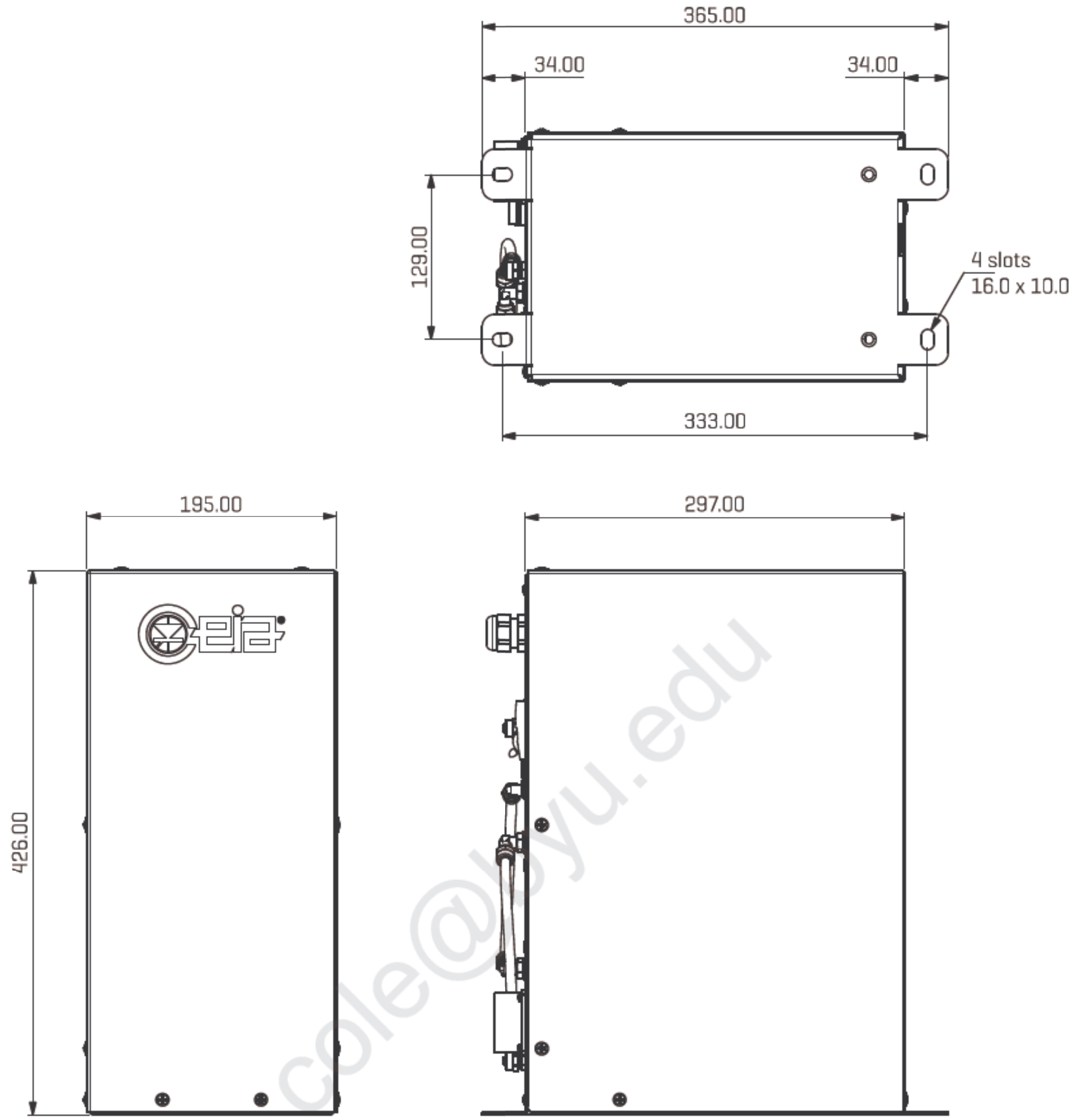
The management of the apparatus is entrusted to a microcomputer that monitors the correct functioning of the machine by means of a self-diagnosis device and signals any necessary interventions on the part of the operator.

The generator is available in several models, which differ in the amount of power delivered and the working frequency:

Model	Working frequency	Power absorbed	Power delivered
Power Cube 90/50	Medium-low frequency	6kW	90 kVAR
Power Cube 90/200	Medium-high frequency		
Power Cube 180/50	Medium-low frequency	12kW	180 kVAR
Power Cube 180/200	Medium-high frequency		



1	*Main switch*
2	*Power supply cable*
3	*The connectors and terminals necessary for the installation of the device*
4	*Connection of the cooling system*
5	*Connection of the heating head*
6	*Label with the indications of the power supply, the model of the equipment and the serial number*

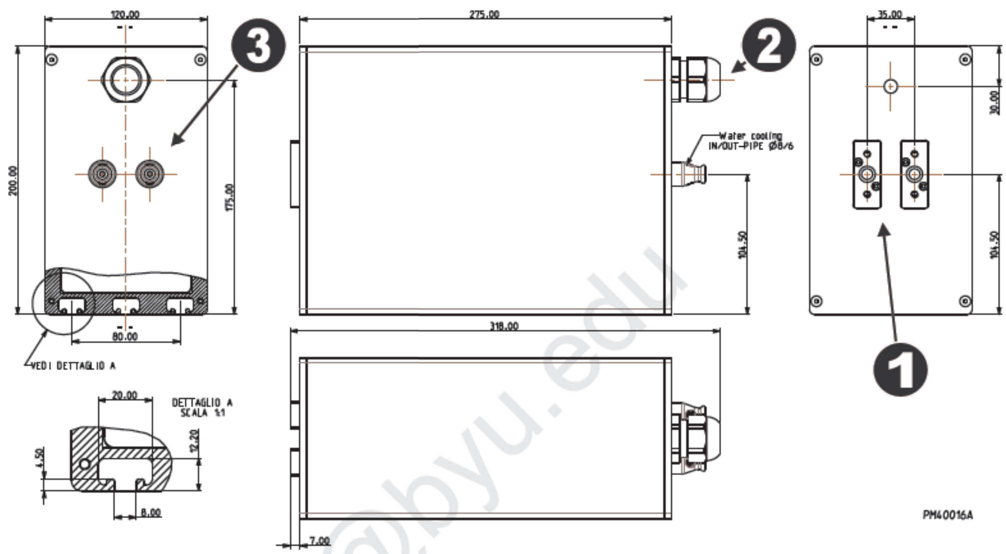

365.00
34.00
34.00
129.00
4 slots
16.0 x 10.0
333.00
195.00
297.00
426.00

2.3.2 Heating head

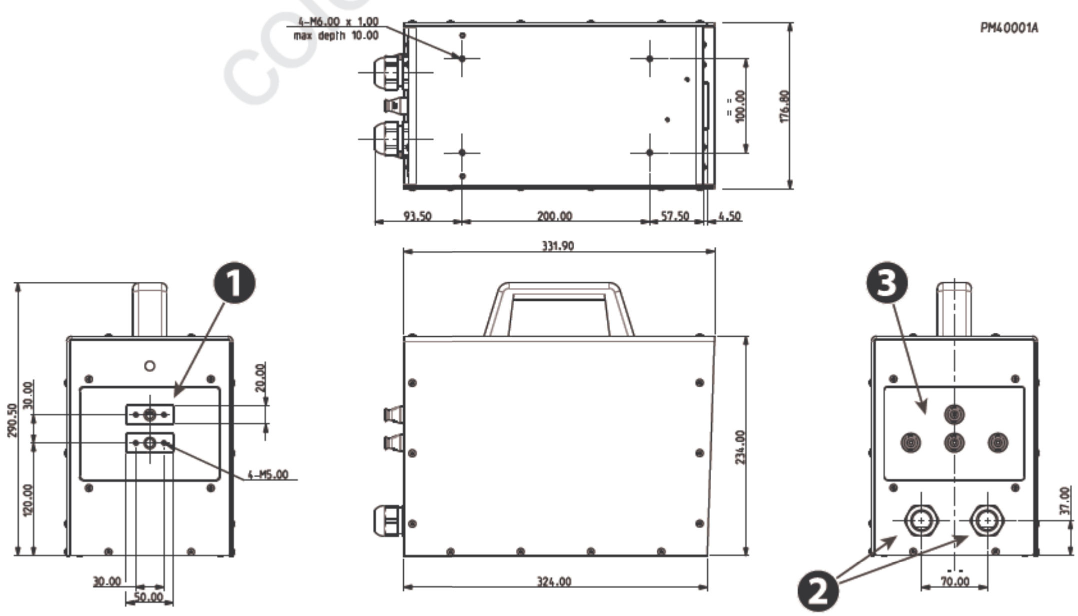
The **PowerCube 90 and 180** generators are designed for the connection with various heating heads:

- models **HH13, HH14, HH16** e **HH23** for PowerCube 90/50 and 180/50;
- models **HH17**, **HH17C, HH18** e **HH23** for PowerCube 90/200 and 180/200.

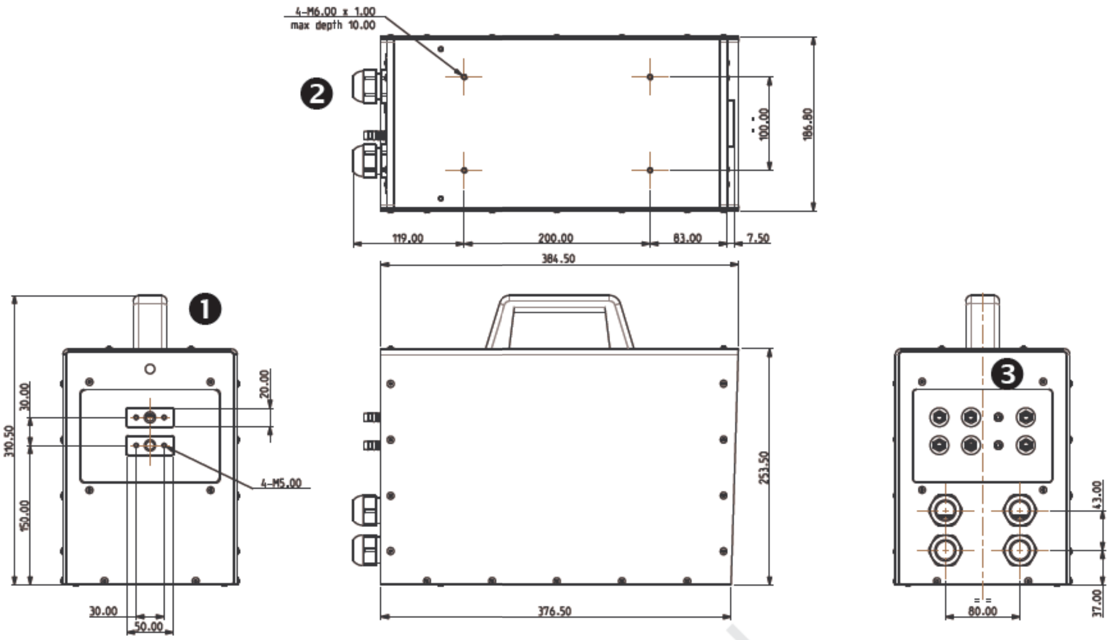
The front panel of the head is fitted with two ❶ metal couplings for fastening the inductor generating the electromagnetic field.
The outlet port for the cables connecting to the generator ❷ and the couplings for the head water cooling ❸ are fitted on the rear side.



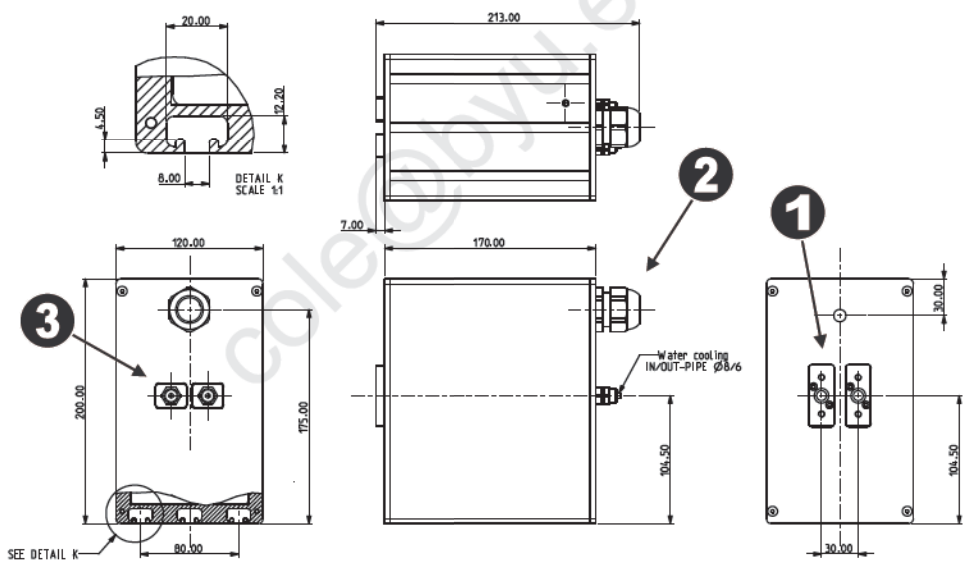
Heating head HH13



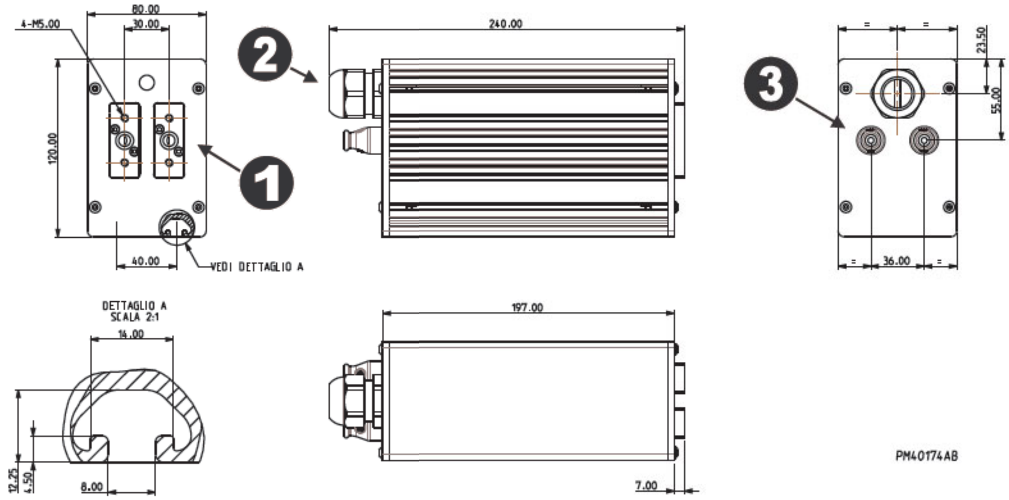
Heating head HH14
(for 24 kW configuration)



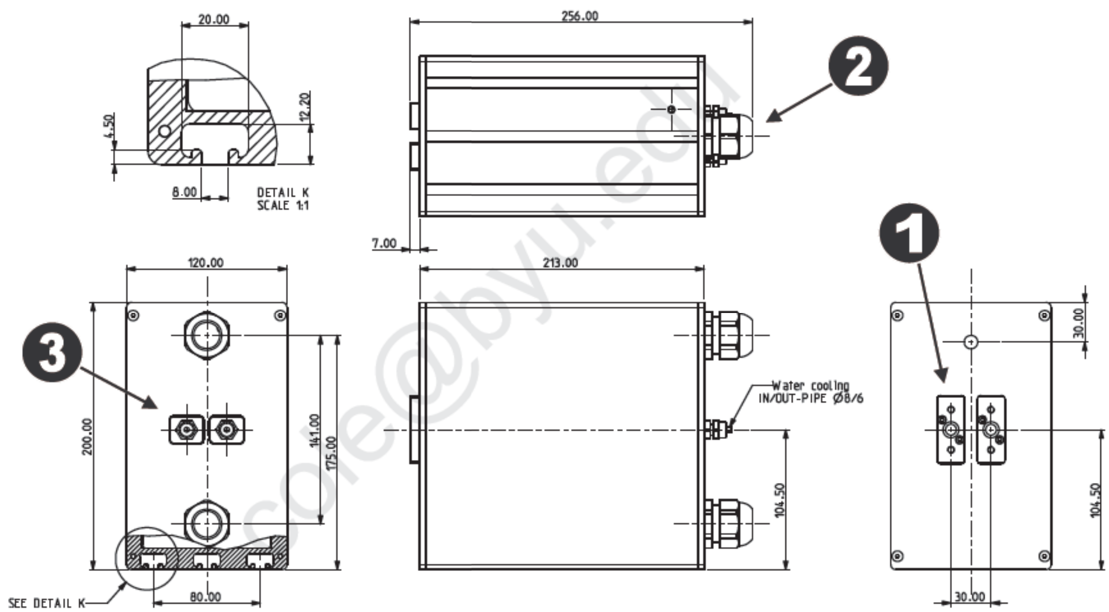
Heating head HH16
(for 48 kW configuration)



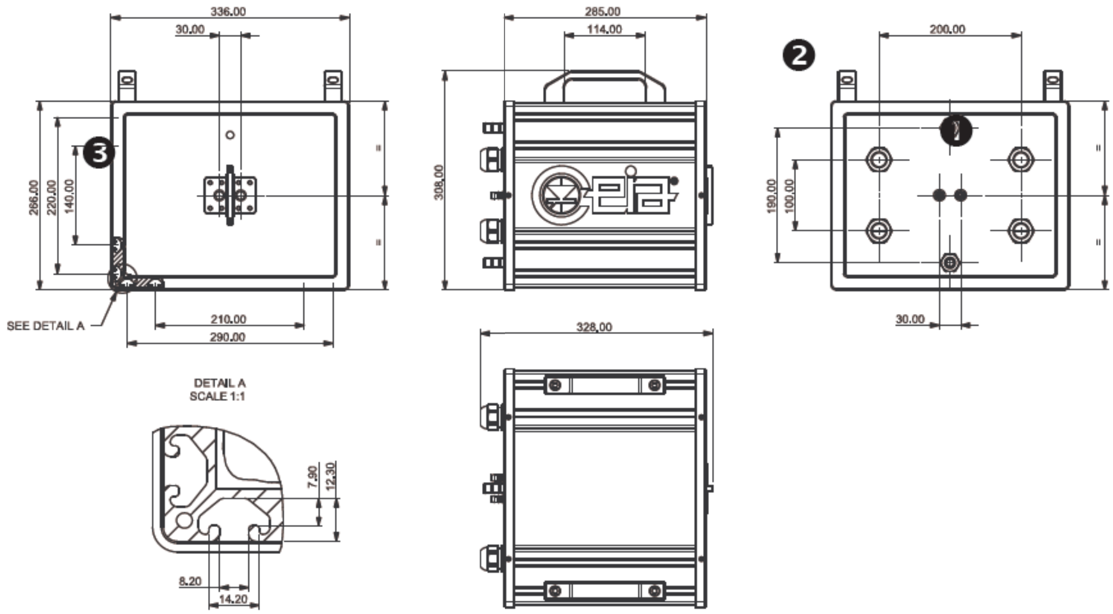
Heating head HH17



Heating head HH17C



Heating head HH18
(for 24 kW configuration)

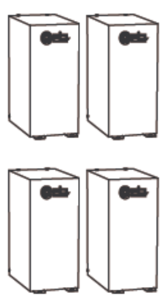


Heating head HH23
(for 48 kW configuration)

2.4 Protection of device and warning signs

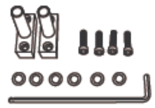
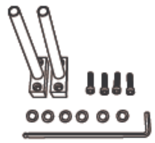
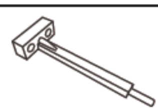
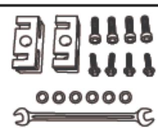
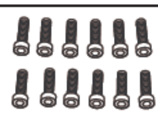
IP Protection Rating	The protection rating of the device is IP20
Self-diagnosis system	Internal. See section 5.2 of this manual.

2.5 Options

Option		Order code
	PowerCube 90/50 Industrial Line	**PW3-90/50**
	PowerCube 90/50 Industrial Line With analogic adjustment of the power	**PW3-90/50-AC**
	PowerCube 90/200 Industrial Line	**PW3-90/200**
	PowerCube 90/200 Industrial Line With analogic adjustment of the power	**PW3-90/200-AC**
	PowerCube 180/50 Industrial Line	**PW3-180/50**
	PowerCube 180/50 Industrial Line With analogic adjustment of the power	**PW3-180/50-AC**
	PowerCube 180/200 Industrial Line	**PW3-180/200**
	PowerCube 180/200 Industrial Line With analogic adjustment of the power	**PW3-180/200-AC**
2 modules in parallel		**Order code**
	Power Cube 180/50 Industrial Line	2x **PW3-360/50-16-**XX
	Power Cube 180/200 Industrial Line	2x **PW3-360/200-13-**XX
4 modules in parallel		**Codice d'ordine**
	Power Cube 180/50 Industrial Line	4x **PW3-720/50-16-**XX
	Power Cube 180/200 Industrial Line	4x **PW3-720/200-13-**XX

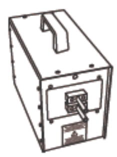
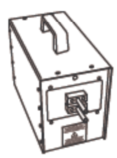
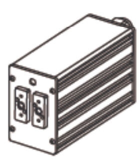
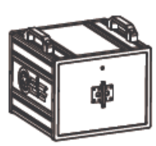
XX = Output impedance coil ratio: 6, 9 o 12

2.6 Accessories

Accessory			Order Code
	Standard inductor holder kit – Series 50 *(2 inductor holders, 6 o-ring, 4 screws, 1 Allen wrench)*		26339
	Standard inductor holder kit – Series 200 *(2 inductor holders, 6 o-ring, 4 screws, 1 Allen wrench)*		30315
	Inductor holder - Series 50 *(1 pc.)*		22648
	Inductor holder - Series 200 *(1 pc.)*		30263
	Inductor holder extension - Series 200		52799
	Thread protection kit		54055
	Inductor holder screws pack *(50 pcs.)*		32841
	Spare o-rings pack *(50 pcs.)*		32840
	Water input filters kit (10 pcs)		74709
	Coil material kit *(copper pipe, vetrotex sheath, tin wire)*	*Copper pipe ø4/3mm x 2m Sheath ø4mm x 2m*	32844
		Copper pipe ø6/4mm x 10m Sheath ø5mm x 10m	32845
		Copper pipe ø8/6mm x 10m Sheath ø8mm x 10m	32846

Accessory			Order Code
	Bundle of copper pipe	*Ø4/3mm x 5m*	74696
		Ø5/4mm x 5m	74697
		Ø6/4mm x 10m	74521
		Ø8/6mm x 10m	74535
		Ø8/6mm x 50m	74663
		Ø10/8mm x 10m	74922
		Ø10/8mm x 50m	74921
	Vetrotex insulation sheath	*Ø4mm x 5m*	74518
		Ø5mm x 5m	74520
		Ø7mm x 10m	74523
		Ø7mm x 50m	74524
		Ø8mm x 10m	74525
		Ø8mm x 50m	74526
		Ø10mm x 10m	74515
		Ø10mm x 50m	74516

2.6.1 Heating Head

Heating Head		For the configuration	Order Code
	**HH13** Heating Head (complete)	12 kW	*XX: available capacitance: 6÷12 uF*
	Cable length 3.0 m		PWH-13-XX-30/50
	Custom cable length		PWH-13-XX-CC/50
	**HH14** Heating Head (complete)	24 kW	*XX: available capacitance: 12÷24 uF*
	Cable length 3.0 m		PWH-14-XX-30/50
	Custom cable length		PWH-14-XX-CC/50
	**HH16** Heating Head (complete)	48 kW	*XX: available capacitance: 12÷24 uF*
	Cable length 3.0 m		PWH-16-XX-30/50
	Custom cable length		PWH-16-XX-CC/50
	**HH17** Heating Head (complete)	12 kW	*XX: available capacitance: 2÷3 uF*
	Cable length 3.0 m		PWH-17-XX-30/200
	Custom cable length		PWH-17-XX-CC/200
	**HH17C** Heating Head (complete)	12 kW	*XX: available capacitance: 1.33÷9.6 uF*
	Cable length 3.0 m		PWH-17-XX-30/200
	Custom cable length		PWH-17-XX-CC/200
	**HH18** Heating Head (complete)	24 kW	*XX: available capacitance: 2÷4 uF*
	Cable length 3.0 m		PWH-18-XX-30/200
	Custom cable length		PWH-18-XX-CC/200
	**HH23** Heating Head (complete)	48 kW	*XX: available capacitance: 2÷5 uF*
	Cable length 3.0 m		PWH-23-XX-30/200
	Custom cable length		PWH-23-XX-CC/200

2.7 Technical features

2.7.1 Features in common

Power supply	Power supply voltage	400V~, 3-phase Δ - 50/60 Hz, without neutral	
Cooling	Water cooling system	With water, using 200 dm³ storage tank with 0,4 Hp pump or with direct off-take at recommended pressure of approx. 300kPa (min. pressure 200kPa, max. pressure 600kPa)	
	Allowed water type	Tap water with PH between 6 and 8; hardness lower than 6°F	
		Deionized or demineralized water, with addition of glycol (with the minimum suggested concentration)	
	Minimum flow rate	1.5 litres/min	
	Water infeed temperature	from ambient temp. to 45°C (without condensation)	
Operating regime	Continuous activation		
Mode of operation	Automatic, driven by one of the management units of the **PowerCube Family**		
Operation and controls	Heating power stabilized automatically (not influenced by power supply voltage variations)		
Self-diagnosis	The fault details are shown on a display and signalled by an acoustic device on the controller or on the digital outputs on the back of the generator	Control of cooling water temperature and pressure	Control of correct design of inductor
		Control of any short-circuit of the heating inductor	Control of the heating head connection
		Internal fault	Control of the power supply voltage value
Generator	Type	Stainless steel container	
	Dimensions (L x W x H)	195 x 365 x 426 mm	
	Power supply cable	3 m	
	Water supply connection for cooling system	Coupling ø1/8" for ø8/6 mm rubber hose	
	Weight	21kg	
Safety features	Galvanic insulation from the mains supply voltage		
	In accordance with current international safety and radio-interference standards and applicable EC directives		
Operating conditions	Operating temperature	+5 / +55 °C	
	Storage temperature	-25 / +70 °C	
	Relative humidity	0 – 95% (without condensation)	

2.7.2 Special features

	PowerCube 90	PowerCube 180
Input current	**8.5A** max Ext. conductors ø 2.5mm² min	**17A** max Ext. conductors ø 4mm² min
Maximum absorbed power	**6 kW**	**12 kW**

2.7.3 Heating heads

	PWH-13	PWH-17	PWH-17C
For use with generator	**Power Cube 90/50** **Power Cube 180/50**	**Power Cube 90/200** **Power Cube 180/200**	**Power Cube 90/50 e /200** **Power Cube 180/50 e /200**
Dimensions (L x W x H) without adductor	120x280x200 mm	120x170x200 mm	80x240x120 mm
Weight	11.5 kg	8.4 kg	7 kg
Capacitance	6-12 uF	2-3 uF	1,33-9,6 uF [1]
Enclosure	High-strength metal casing		
Inductors with shape corresponding to part being processed	The use of a copper tube (minimum diameter ø 8/6 mm) is recommended for the construction	The use of a copper tube (minimum diameter ø 4/3 mm) is recommended for the construction	
Cable for the connection with generator	3 m length (standard)		
Water cooling connection with the generator	Coupling ø1/8", rubber hose ø8/6 mm		

	PWH-14	PWH-18
For use with generator	**Power Cube 180/50** – 2 units	**Power Cube 180/200** – 2 units
Dimensions (L x W x H) without adductor	177x234x332 mm	120x213x200 mm
Weight	20 kg	13 kg
Capacitance	12-24 uF	2-4 uF
Enclosure	High-strength metal casing	
Inductors with shape corresponding to part being processed	The use of a copper tube (minimum diameter ø8/6 mm) is recommended for the construction	The use of a copper tube (minimum diameter ø4/3 mm) is recommended for the construction
Cable for the connection with generator	3 m length (standard)	
Water cooling connection with the generator	Coupling ø1/8", rubber hose ø8/6 mm	

	PWH-16	PWH-23
For use with generator	**Power Cube 180/50** – 4 units	**Power Cube 180/200** – 4 units
Dimensions (L x W x H) without adductor	187x253,5x384,5 mm	336x285x266 mm
Weight	37 kg	30 kg
Capacitance	12-24 uF	2-5 uF
Enclosure	High-strength metal casing	
Inductors with shape corresponding to part being processed	The use of a copper tube (minimum diameter ø10/8 mm) is recommended for the construction	The use of a copper tube (minimum diameter ø6/4 mm) is recommended for the construction
Cable for the connection with generator	3 m length (standard)	
Water cooling connection with the generator	ø17/8 mm rubber pipe barb	ø17/12 mm rubber pipe barb

[1] *Depending on the frequency of the Generator*

3 INSTALLATION

Read the Section **Safety Instructions – Warnings** in this manual before carrying out the machine installation operations.

3.1 Mechanical installation

3.1.1 Minimum requirements

Position and attach the welding station components in accordance with the processing requirements, to allow installation in consideration of:

- length of connection cables;
- electricity and water supply requirements;
- fixing methods.

Set up the workstation by installing the parts in order to make the processing easy.

3.1.2 Assembly

The **PowerCube** is supplied completely assembled.

3.1.3 Transport

WARNING! Disconnect the system from all supply sources before performing any movement.

Take care when moving the equipment, as it may be too heavy to be handled by only one person. Set up and implement adequate procedures for performing this operation. The supplier may not be held responsible for injury to persons or damage to property resulting from incorrect handling and transport.

3.1.4 Connection to the cooling system

Refer to the Technical Features for the dimensions of the system.

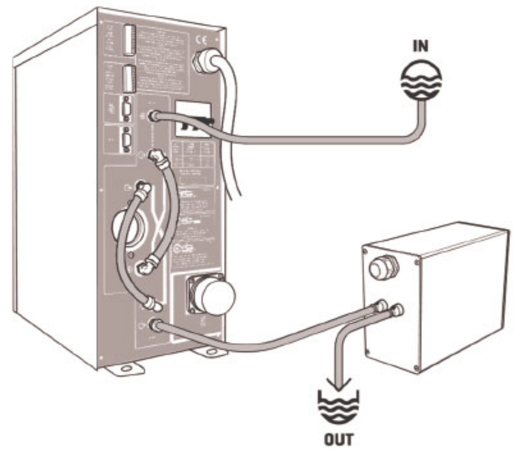
WARNING! It is absolutely important that the cooling water inside the generator is at a temperature between the environment temperature and a maximum of 45°C, without condensation.

WARNING! Do not invert the inlet and outlet.

IMPORTANT! - Cooling water flow rate
To ensure correct operation, the flow rate must not be less than a minimum value, shown in the “Technical features” for each model. The use of inductors made from pipes with small cross-sections or with layouts which cause restrictions (very small dimensions or sharp bends) may cause a reduction in the flow rate, such as to result in overheating of the inductor. If the processing needs require the use of such inductors, it is necessary to increase the water supply pressure, whilst remaining within the limits indicated in “Technical features”.

3.1.4.1 Configuration with single generator

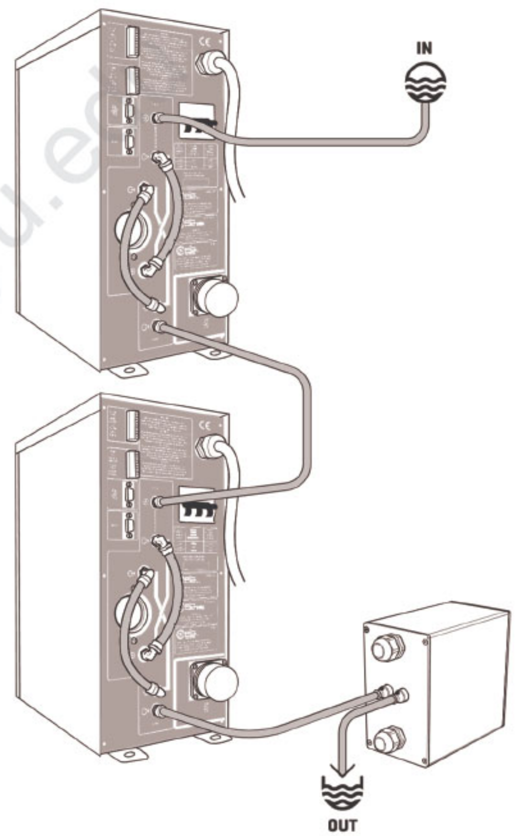
- *Connect the output of the external cooling unit to the **H_2O IN** terminal of the generator; insert a security tap up-stream.*
- *Connect the input of the external cooling unit to the **H_2O OUT** terminal of the generator.*
- *Connect the return of the cooling unit to the fitting left free on the heating head.*



Example with a HH13 heating head

3.1.4.2 Configuration with two generators

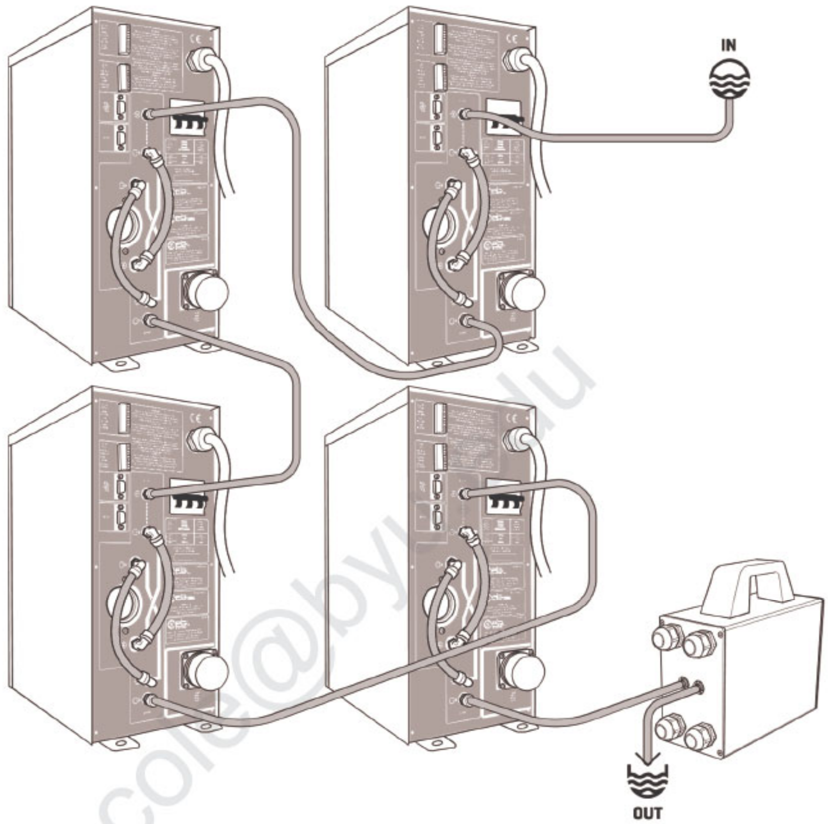
- *Connect the output of the external cooling unit to the **H_2O IN** terminal of the generator; insert a security tap up-stream.*
- Connect the **H_2O OUT** output of the main generator to the **H_2O IN** input of the secondary generator.
- *Connect the input of the external cooling unit to the **H_2O OUT** terminal of the secondary generator.*
- *Connect the return of the cooling unit to the fitting left free on the heating head.*



Example with a HH18 heating head

3.1.4.3 Configuration with four generators

- Connect the output of the external cooling unit to the **H_2O IN** terminal of the generator; insert a security tap up-stream.
- Connect in cascade the **H_2O OUT** of each generator to the **H_2O IN** of the next generator.
- Connect the input of the external cooling unit to the **H_2O OUT** terminal of the last secondary generator.
- Connect the return of the cooling unit to the fitting left free on the heating head.



Example with a HH16 heating head

3.2 Electrical installation

3.2.1 Minimum requirements

Ref. values in subsection *2.7 - Technical features.*

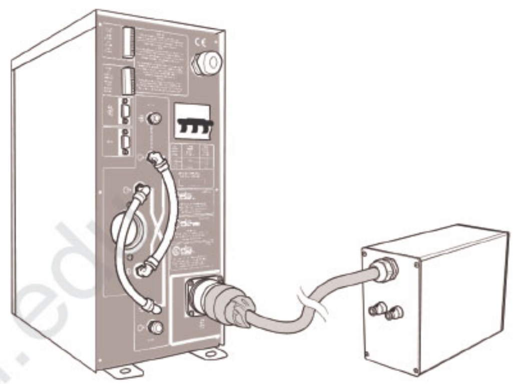
3.2.2 Selection and installation of connection cables

Power Cube 90 e 180 is supplied complete with all the cables necessary for operation.
Do not use different cables.

3.2.3 Connection of the heating heads

3.2.3.1 Configuration with single generator

- Connect the head cable to the generator plug.
- Securely fasten the connector fixing nut.

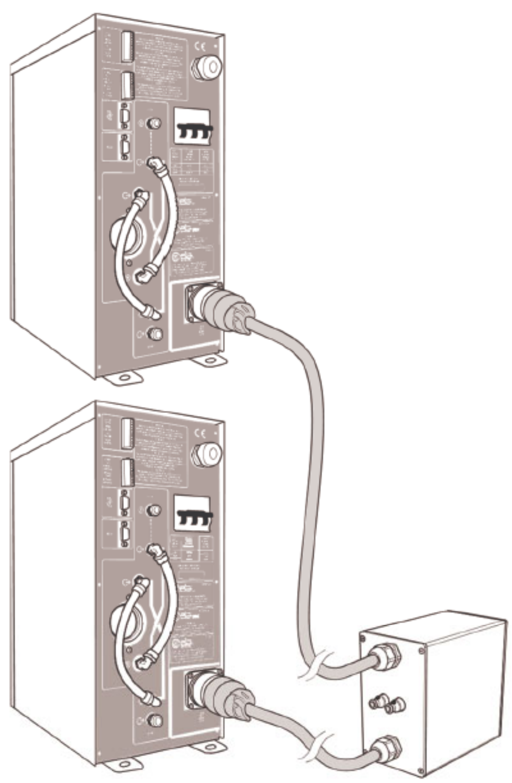


Example with a HH13 heating head

3.2.3.2 Configuration with two generators

Only modules of the same type can be synchronized.

- Connect the head cables to the plugs of the two generators.
- Securely fasten the connector fixing nuts.

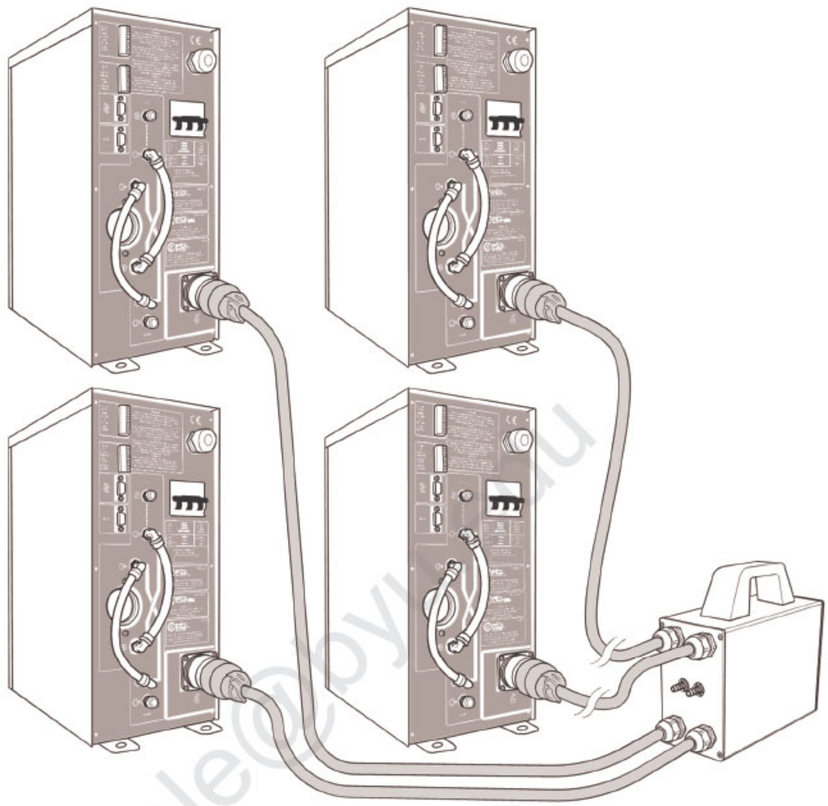


Example with a HH18 heating head

3.2.3.3 Configuration with four generators

Only modules of the same type can be synchronized.

- Connect the head cables to the plugs of the four generators.
- Securely fasten the connector fixing nuts.

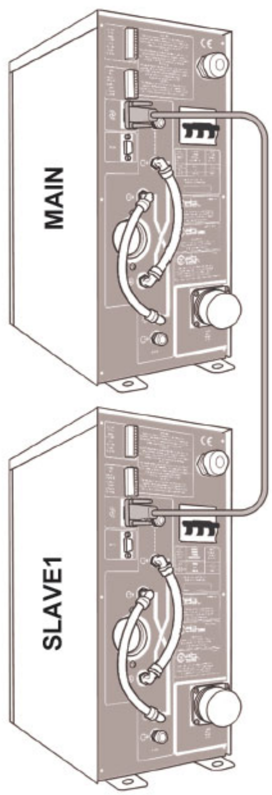


Example with a HH16 heating head

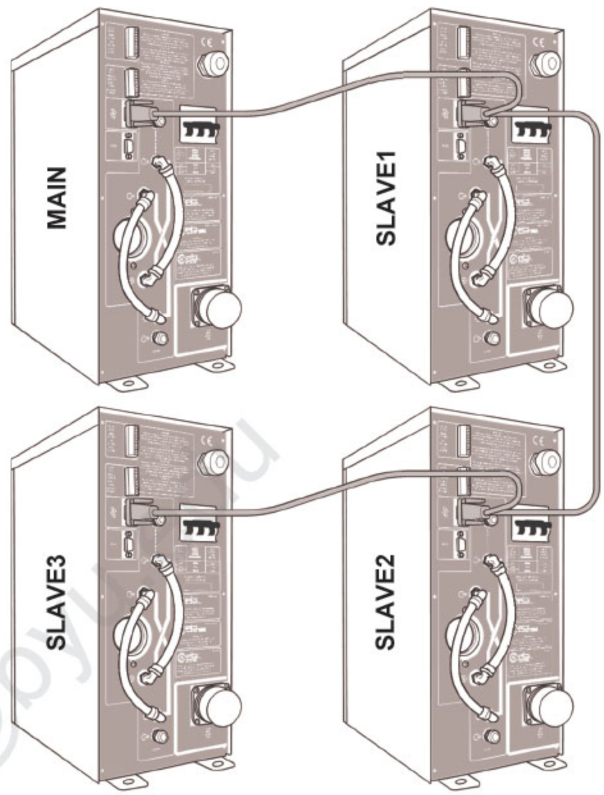
3.2.4 Connection of the synchronisation cable

Only modules of the same type can be synchronized.

In configurations with two or four generators, the synchronisation cable between the different generators must also be connected, as shown below.



Connection of two generators via synchronisation cable 22907



Connection of four generators via synchronisation cable 22909

Each connector on the synchronisation cable shows the operating mode of the module to which it is connected (MAIN, SLAVE1, SLAVE2, SLAVE3).

3.2.5 Connection to mains power supply

Only connect the device to the mains supply after making all the connections for the complete installation of the equipment.

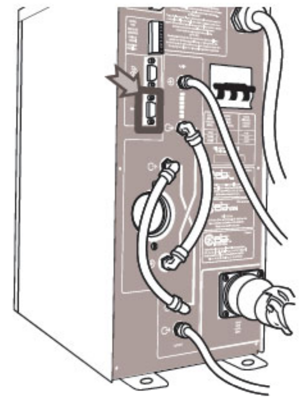
Connect the machine to earth and to the mains voltage specified on the nameplate, using the relative power supply cable.

The external line must have conductors with a **cross-section suitable for the current absorbed** (see section 2.7).

3.2.6 Connection with other PowerCube Family devices

For the connection of other **PowerCube Family** devices, like the **Master Controller**, use the **RS 232** connector (female 9-poles sub D type), located on the back.

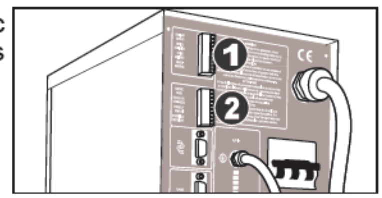
> Except for the pin 9, that have a 24Vcc max voltage, the connector works as a normal RS-232 interface.

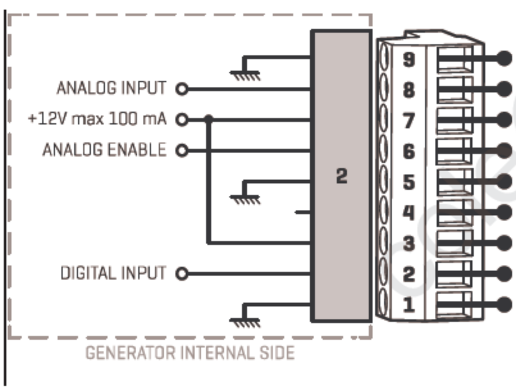


Pin	Function
2	RXD
3	TXD
5	SIGNAL GND
9	+24Vdc, 150mA max

3.2.7 Connection for the remote analogic adjustment of the power (only on PowerCubeXX-AC)

The connection with the external management unit, for the analogic adjustment of the power, is performed through the 9-poles connector CONTROL PANEL, located on the back.





Pin	Type	Function	Range
1	GND	-	-
2	Input	Generator activation	0/5Vdc (max 0÷30Vdc)
3	+12V	-	Imax 100mA [1]
4	-	Reserved	-
5	GND	-	-
6	Input	Analog input enable	0/5Vdc (max 0÷30Vdc)
7	Out +12V	-	Imax 100mA [1]
8	Input	Analog input power regulation	IN 0÷10V Zin=100KΩ (max -24÷+24 Vdc)
9	GND	-	-

Maximum length of the cable: 3 m (Recommended cable type: shielded cable)

4 USE

4.1 Use of the equipment

Before using the machine, read the Section Safety Instructions – Warnings in this manual. It should be noted that CEIA may not be held responsible for any damage due to failure to comply with the requirements specified in this manual.

The **PowerCube** generator can be the generating part of a heating workstation controlled by a **MasterController**, which manages all the devices required in a heating cycle, such as, for example, anti-oxidising gas diffusers, temperature sensors and soldered-wire feeders.
The **Power Cube** itself is also fully controlled by the device.
For information about the equipment in the **Power Cube Family**, contact our sales department.

The list of operations that follows assumes that the inductor, any equipment for positioning the piece to be processed and any accessory devices have already been manufactured and properly positioned during the heating system installation phase, so that the operator can carry out the work cycle correctly and safely.

Before the equipment is put into operation all the safety and protection devices aimed at preventing parts of the body or objects from coming close to the heating inductor during activation of the generator must be set up.

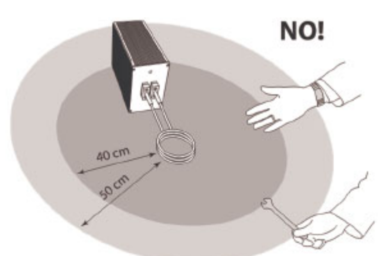
1. Provide the necessary electricity and water supply.
   Operate the ON-OFF switch installed on the front of the generator.

2. Program the desired power values on the controller, if needed.

Before the equipment is put into operation all the safety and protection devices aimed at preventing parts of the body or objects from coming close to the heating inductor during activation of the generator must be set up.

3. Position the parts to be welded close to the heating inductor, **taking care not to come into contact with the latter**.

In particular, direct or indirect contact with the heating inductor must be prevented and a minimum safety distance **SD** must be maintained, depending on the type of inductor used, of up to **80 cm**.
The heating of metal objects may occur below these clearance distances, even without direct contact, with the risk of burning for the operator.



During activation of the generator the heating inductor is supplied with a dangerous voltage (in accordance with CEI EN 61010-1, subsection 6.3.1) : avoid direct contact or indirect contact with metal objects.

INTENSE ELECTROMAGNETIC FIELDS may have an affect on pacemakers. Persons wearing pacemakers, cardiac defibrillators and other support devices must not operate close to the inductor or the heating inductors of the machine. These individuals must consult their personal doctor if they are to operate other than as stated above.

4. Activate the heating cycle and carry out the necessary operations.

During the generator activation, the heating inductor and the piece being worked reach very high temperature: avoid any direct contact, even after the generator deactivation, to avoid any risk of burning.

5. At the end of the process remove the last piece processed from the inductor, switch OFF the electricity supply and close the water supply valve.

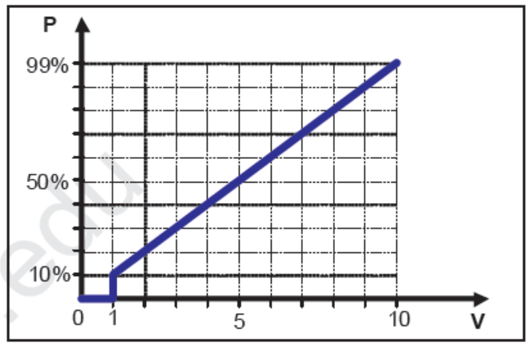
4.2 Programming

For all the information about programming operations, refers to the Management Unit manual.

4.2.1 Power adjustment (only on PowerCubeXX-AC)

Power / voltage ratio

The output is proportional to the voltage at the control input, in the range 10%…99%. When the voltage control input is less than 1 V, the generator is deactivated.



5 MAINTENANCE

Read the Section **Safety Instructions – Warnings** in this manual before carrying out any maintenance operations on the machine.

5.1 Regular maintenance

The device does not require periodic maintenance.

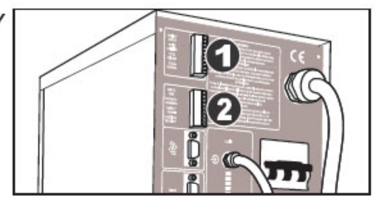
We recommend in any case that the components of the system, especially the inductor, are checked before operation.

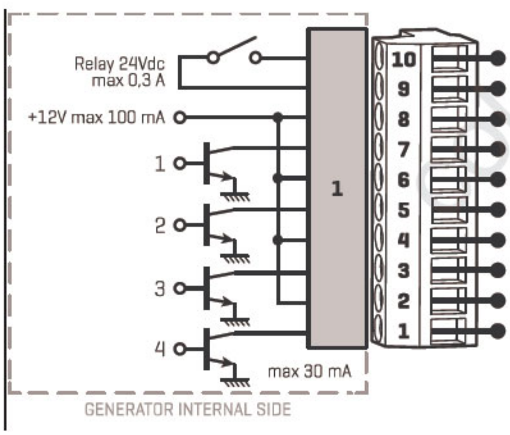
The operator must assess the condition of the system and its suitability for use, if necessary consulting the maintenance service.

5.2 Internal self-diagnosis

The self-diagnosis device of the **PowerCube 90 and 180** warns the operator, in case of fault, through a series of self-diagnosis messages shown on the controller display and through an acoustical alarm.

These messages are also shown on the 10-pin AUXILIARY OUTPUT connector on the rear.





Pin	Type	Function	Range
1	Open Collector	Out 4 diagnosis	Imax 30mA
2	Out +12V	-	Imax 100mA [2]
3	Open Collector	Out 3 diagnosis	Imax 30mA
4	Out +12V	-	Imax 100mA [1]
5	Open Collector	Out 2 diagnosis	Imax 30mA
6	Out +12V	-	Imax 100mA [1]
7	Open Collector	Out 1 diagnosis	Imax 30mA
8	Out +12V	-	Imax 100mA [1]
9 10	Out relé N.A.	Generator ready (relay closed)	Vmax 48Vdc, Imax 0,5A

[2] *The total current that can be drawn at the +12V terminals is 100mA.*

5.2.1 Self-diagnosis messages

Code	Message	Signalling on Open Collector output				Possible cause	Possible action
		1	2	3	4		
-	No diagnosis	OFF	OFF	OFF	OFF	-	-
A1	Water temperature too high	OFF	OFF	OFF	**ON**	Cooling circuit malfunction	Check the cooling water temperature. It must be below 50°C. (Trigger point = 55°C)
A2	Internal fault	OFF	**ON**	**ON**	OFF	Internal fault	Contact the CEIA Technical Assistance Service
A3[2]	No cooling water	OFF	OFF	**ON**	**ON**	Insufficient flow of cooling water	Check that there is water in the cooling circuit. Check the presence of blocking-up in the generator inlets and outlets.
A4	Check short circuit in adductors	OFF	**ON**	OFF	OFF	Short circuit between the inductor lobes	Check for a short circuit in the inductors
	Increase coil dimension					Abductor dimensions are too small	Increase the coil dimensions
A5	Decrease coil dimension	OFF	**ON**	OFF	**ON**	Too big dimensions of the inductor	Decrease the dimensions of the abductor
A6[2]	Internal fault	OFF	**ON**	**ON**	OFF	Fault in the power circuit	Contact the CEIA Technical Assistance Service.
A7	Head not connected	OFF	**ON**	**ON**	**ON**	Heating head disconnected	Check the connection between the heads and the electronics unit
A10	Low line voltage	**ON**	OFF	**ON**	OFF		Insert (between the mains and the heater) a transformer with an input voltage equal to that available on the line and an output voltage of 220Vac
A11	High line voltage	**ON**	OFF	**ON**	**ON**		
A12	Generator not ready	**ON**	**ON**	OFF	OFF	One of the three phases of the line is not connected	Check the connection to the electrical system
A21	SLAVE1 not found	OFF	**ON**	**ON**	OFF	One of the SUB generators is not connected correctly or is switched off	Check that the supply voltage is present and that the relevant SUB generator is switched on correctly. Check the connection of the synchronization cable between the generators
A22	SLAVE2 not found						
A23	SLAVE3 not found						

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6 APPENDIX

6.1 Declaration of Conformity CE

6.1.1 PowerCube 90/50

EU DECLARATION OF CONFORMITY

(in accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSERKLÄRUNG (nach EN ISO/IEC 17050-1:2010)
DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010)

This declaration of conformity is issued under the sole responsibility of the manufacturer */ La présente déclaration de conformité est établie sous la seule responsabilité du fabricant / Die alleinige Verantwortung für die Ausstellung dieser Konformitätserklärung trägt der Hersteller / La presente declaración de conformidad se expide bajo la exclusiva responsabilidad del fabricante / La presente dichiarazione di conformità è rilasciata sotto la responsabilità esclusiva del fabbricante*

Manufacturer: **_Fabricant_** */ Hersteller / Fabricante / Costruttore:*	**_CEIA S.p.A._** *Zona industriale Viciomaggio 54* *52041 Viciomaggio Arezzo – ITALY*

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: *Nom du produit: / Produktname:* *Nombre del producto: / Nome:*	**_Inductive Heater_** *Réchauffeur de métaux a induction / Induktions-Lötgeräte* *Calentador por inducción / Riscaldatore induttivo di metalli*
Model: *Modelé / Modell / Modelo / Modello:*	**_PW3-90/50_**
Product Options: *Options / options:* *Opciones / opzioni:*	**_This declaration covers all options_** *Cette déclaration est valide pour toutes les options / Diese Erklärung ist gültig für alle options* *Este declaración es valida para todas las opciones / Questa dichiarazione è valida per tutte le opzioni*

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

_EN 60204-1:2018_ *Safety of machinery - Electrical equipment of machines – Part 1: General requirements*

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.
Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.
Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU.
El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.
Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 55011:2016 + EN 55011:2016/A1:2017 + EN 55011:2016/A11:2020 Group 2 - Class A *Industrial, scientific and medical (ISM) radio-frequency equipment - Electromagnetic disturbance characteristics - Limits and methods of measurement*
_EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005_ *Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments*

This product complies with the requirements of the EMC Directive 2014/30/EU.
Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.
Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.
El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.
Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The device is designed to be inserted as a component part of a machine; the safety devices needed to protect the operators must therefore be designed and fitted by the installer according to the type of application in question.
L'appareil a été conçu pour faire partie d'une machine; cependant, les dispositifs de sécurité nécessaires à la protection des opérateurs devront être conçus et réalisés par le monteur en fonction du type d'application même.
Das Gerät wurde entwickelt, um als Bestandteil in eine Fertigungsanlage integriert zu werden; die zum Schutz des Personals erforderlichen Sicherheitsvorrichtungen müssen deshalb durch den Installateur je nach der Anwendungsart überarbeitet und realisiert werden.
El equipo esta diseñado para ser insertado como un componente de una máquina; las protecciones necesarias para la seguridad del operario deben de ser diseñadas y situadas por el instalador de acuerdo al tipo de trabajo en cuestión.
L'apparato è progettato per l'inserimento come componente di una macchina; pertanto, i dispositivi di sicurezza necessari per la protezione degli operatori devono essere progettati e realizzati dall'installatore, in funzione del tipo di applicazione.

Signed for and on behalf of: */ Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di:*
CEIA S.p.A. *Zona industriale Viciomaggio 54 52041 Viciomaggio Arezzo – ITALY*

Arezzo, 2021-02-22

SpA

Ing. Enrico Sorini

Lab. EMC Person in charge
Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor
Resp. Laboratorio EMC / Resp. Lab. EMC

Doc. N° PW3-90/50 rev 4 Mod. P0401-0025 rev 0

6.1.2 PowerCube 180/50

EU DECLARATION OF CONFORMITY

(in accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSERKLÄRUNG (nach EN ISO/IEC 17050-1:2010)
DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010)

This declaration of conformity is issued under the sole responsibility of the manufacturer */ La présente déclaration de conformité est établie sous la seule responsabilité du fabricant / Die alleinige Verantwortung für die Ausstellung dieser Konformitätserklärung trägt der Hersteller / La presente declaración de conformidad se expide bajo la exclusiva responsabilidad del fabricante / La presente dichiarazione di conformità è rilasciata sotto la responsabilità esclusiva del fabbricante*

Manufacturer: ***Fabricant*** */ Hersteller / Fabricante / Costruttore:*	***CEIA S.p.A.*** *Zona industriale Viciomaggio 54* *52041 Viciomaggio Arezzo – ITALY*

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: *Nom du produit: / Produktname:* *Nombre del producto: / Nome:*	***Inductive Heater*** *Réchauffeur de métaux a induction / Induktions-Lötgeräte* *Calentador por inducción / Riscaldatore induttivo di metalli*
Model: *Modelé / Modell / Modelo / Modello:*	***PW3-180/50***
Product Options: *Options / options:* *Opciones / opzioni:*	***This declaration covers all options*** *Cette déclaration est valide pour toutes les options / Diese Erklärung ist gültig für alle options* *Este declaración es valida para todas las opciones / Questa dichiarazione è valida per tutte le opzioni*

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

EN 60204-1:2018 *Safety of machinery - Electrical equipment of machines – Part 1: General requirements*

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.
Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.
Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU.
El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.
Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 55011:2016 + EN 55011:2016/A1:2017 + EN 55011:2016/A11:2020 Group 2 - Class A *Industrial, scientific and medical (ISM) radio-frequency equipment - Electromagnetic disturbance characteristics - Limits and methods of measurement*
EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005 *Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments*

This product complies with the requirements of the EMC Directive 2014/30/EU.
Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.
Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.
El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.
Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The device is designed to be inserted as a component part of a machine; the safety devices needed to protect the operators must therefore be designed and fitted by the installer according to the type of application in question.
L'appareil a été conçu pour faire partie d'une machine; cependant, les dispositifs de sécurité nécessaires à la protection des opérateurs devront être conçus et réalisés par le monteur en fonction du type d'application même.
Das Gerät wurde entwickelt, um als Bestandteil in eine Fertigungsanlage integriert zu werden; die zum Schutz des Personals erforderlichen Sicherheitsvorrichtungen müssen deshalb durch den Installateur je nach der Anwendungsart überarbeitet und realisiert werden.
El equipo esta diseñado para ser insertado como un componente de una máquina; las protecciones necesarias para la seguridad del operario deben de ser diseñadas y situadas por el instalador de acuerdo al tipo de trabajo en cuestión.
L'apparato è progettato per l'inserimento come componente di una macchina; pertanto, i dispositivi di sicurezza necessari per la protezione degli operatori devono essere progettati e realizzati dall'installatore, in funzione del tipo di applicazione.

Signed for and on behalf of: */ Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di:*
CEIA S.p.A. *Zona industriale Viciomaggio 54 52041 Viciomaggio Arezzo – ITALY*

Arezzo, 2021-02-22

SpA

Ing. Enrico Sorini

Lab. EMC Person in charge
Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor
Resp. Laboratorio EMC / Resp. Lab. EMC

Doc. N° PW3-180/50 rev 4 Mod. P0401-0025 rev 0

6.1.3 PowerCube 90/200

EU DECLARATION OF CONFORMITY

(in accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSERKLÄRUNG (nach EN ISO/IEC 17050-1:2010)
DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÁ UE (in conformità con EN ISO/IEC 17050-1:2010)

This declaration of conformity is issued under the sole responsibility of the manufacturer */ La présente déclaration de conformité est établie sous la seule responsabilité du fabricant / Die alleinige Verantwortung für die Ausstellung dieser Konformitätserklärung trägt der Hersteller / La presente declaración de conformidad se expide bajo la exclusiva responsabilidad del fabricante / La presente dichiarazione di conformità è rilasciata sotto la responsabilità esclusiva del fabbricante*

Manufacturer: ***Fabricant*** */ Hersteller / Fabricante / Costruttore:*	***CEIA S.p.A.*** *Zona industriale Viciomaggio 54* *52041 Viciomaggio Arezzo – ITALY*

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: *Nom du produit: / Produktname:* *Nombre del producto: / Nome:*	***Inductive Heater*** *Réchauffeur de métaux a induction / Induktions-Lötgeräte* *Calentador por inducción / Riscaldatore induttivo di metalli*
Model: *Modelé / Modell / Modelo / Modello:*	***PW3-90/200***
Product Options: *Options / options:* *Opciones / opzioni:*	***This declaration covers all options*** *Cette déclaration est valide pour toutes les options / Diese Erklärung ist gültig für alle options* *Este declaración es valida para todas las opciones / Questa dichiarazione è valida per tutte le opzioni*

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

EN 60204-1:2018 *Safety of machinery - Electrical equipment of machines – Part 1: General requirements*

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.
Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.
Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU.
El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.
Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 55011:2016 + EN 55011:2016/A1:2017 + EN 55011:2016/A11:2020 Group 2 - Class A *Industrial, scientific and medical (ISM) radio-frequency equipment - Electromagnetic disturbance characteristics - Limits and methods of measurement*
EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005 *Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments*

This product complies with the requirements of the EMC Directive 2014/30/EU.
Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.
Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.
El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.
Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The device is designed to be inserted as a component part of a machine; the safety devices needed to protect the operators must therefore be designed and fitted by the installer according to the type of application in question.
L'appareil a été conçu pour faire partie d'une machine; cependant, les dispositifs de sécurité nécessaires à la protection des opérateurs devront être conçus et réalisés par le monteur en fonction du type d'application même.
Das Gerät wurde entwickelt, um als Bestandteil in eine Fertigungsanlage integriert zu werden; die zum Schutz des Personals erforderlichen Sicherheitsvorrichtungen müssen deshalb durch den Installateur je nach der Anwendungsart überarbeitet und realisiert werden.
El equipo esta diseñado para ser insertado como un componente de una máquina; las protecciones necesarias para la seguridad del operario deben de ser diseñadas y situadas por el instalador de acuerdo al tipo de trabajo en cuestión.
L'apparato è progettato per l'inserimento come componente di una macchina; pertanto, i dispositivi di sicurezza necessari per la protezione degli operatori devono essere progettati e realizzati dall'installatore, in funzione del tipo di applicazione.

Signed for and on behalf of: */ Signé par et au nom de: / Unterzeichnet für und im Namen von: / Firmado por y en nombre de: / Firmato in vece e per conto di:*
CEIA S.p.A. *Zona industriale Viciomaggio 54 52041 Viciomaggio Arezzo – ITALY*

Arezzo, 2021-02-22

SpA

Ing. Enrico Sorini

Enrico Sorini

Lab. EMC Person in charge
Resp. Laboratoire EMC / Verantwortlicher für EMC-Labor
Resp. Laboratorio EMC / Resp. Lab. EMC

Doc. N° PW3-90/200 rev 4 Mod. P0401-0025 rev 0

6.1.4 PowerCube 180/200

EU DECLARATION OF CONFORMITY

(in accordance with EN ISO/IEC 17050-1:2010)

DECLARATION DE CONFORMITE UE (conformément à la norme EN ISO/IEC 17050-1:2010) EU-KONFORMITÄTSERKLÄRUNG (nach EN ISO/IEC 17050-1:2010)
DECLARACION DE CONFORMIDAD UE (según EN ISO/IEC 17050-1:2010) DICHIARAZIONE DI CONFORMITÀ UE (in conformità con EN ISO/IEC 17050-1:2010)

This declaration of conformity is issued under the sole responsibility of the manufacturer */ La présente déclaration de conformité est établie sous la seule responsabilité du fabricant / Die alleinige Verantwortung für die Ausstellung dieser Konformitätserklärung trägt der Hersteller / La presente declaración de conformidad se expide bajo la exclusiva responsabilidad del fabricante / La presente dichiarazione di conformità è rilasciata sotto la responsabilità esclusiva del fabbricante*

Manufacturer: **Fabricant** / *Hersteller / Fabricante / Costruttore:*	***CEIA S.p.A.*** *Zona industriale Viciomaggio 54* *52041 Viciomaggio Arezzo – ITALY*

Declares that the product

déclare que ce produit / erklärt, daß das Produkt / declara que el producto / dichiara che il prodotto:

Product name: *Nom du produit: / Produktname: Nombre del producto: / Nome:*	***Inductive Heater*** *Réchauffeur de métaux a induction / Induktions-Lötgeräte* *Calentador por inducción / Riscaldatore induttivo di metalli*
Model: *Modelé / Modell / Modelo / Modello:*	***PW3-180/200***
Product Options: *Options / options: Opciones / opzioni:*	***This declaration covers all options*** *Cette déclaration est valide pour toutes les options / Diese Erklärung ist gültig für alle options* *Este declaración es valida para todas las opciones / Questa dichiarazione è valida per tutte le opzioni*

conforms to the following Product Specifications

est conforme aux spécifications suivantes / folgenden Produktspezifikationen entspricht
es conforme a las siguientes especificaciones / è conforme alle seguenti specifiche di prodotto:

Low Voltage (LVD) Directive 2014/35/EU

EN 60204-1:2018 *Safety of machinery - Electrical equipment of machines – Part 1: General requirements*

This product complies with the requirements of the Low Voltage Directive 2014/35/EU.
Le produit ci-dessus répond aux exigences de la Directive 2014/35/UE concernant la basse tension.
Dieses Produkt entspricht den Anforderungen an Niederspannungsgeräte gemäß der Norm 2014/35/EU.
El producto indicado cumple los requisitos de la Low Voltage Directive 2014/35/UE.
Il prodotto è conforme alle norme della direttiva 2014/35/UE sulla bassa tensione.

Electromagnetic Compatibility (EMC) Directive 2014/30/EU

EN 55011:2016 + EN 55011:2016/A1:2017 + EN 55011:2016/A11:2020 Group 2 - Class A *Industrial, scientific and medical (ISM) radio-frequency equipment - Electromagnetic disturbance characteristics - Limits and methods of measurement*
EN 61000-6-2:2005 + EN 61000-6-2:2005/AC:2005 *Electromagnetic compatibility (EMC) -- Part 6-2: Generic standards - Immunity for industrial environments*

This product complies with the requirements of the EMC Directive 2014/30/EU.
Le produit ci-dessus répond aux exigences de la Directive 2014/30/UE concernant les interférences électromagnétiques.
Dieses Produkt entspricht den Anforderungen der EMC-Norm 2014/30/EU.
El producto indicado cumple los requisitos de la Directiva EMC 2014/30/UE.
Il prodotto è conforme alle norme della direttiva EMC 2014/30/UE.

The device is designed to be inserted as a component part of a machine; the safety devices needed to protect the operators must therefore be designed and fitted by the installer according to the type of application in question.
L'appareil a été conçu pour faire partie d'une machine; cependant, les dispositifs de sécurité nécessaires à la protection des opérateurs devront être conçus et réalisés par le monteur en fonction du type d'application même.
Das Gerät wurde entwickelt, um als Bestandteil in eine Fertigungsanlage integriert zu werden; die zum Schutz des Personals erforderlichen Sicherheitsvorrichtungen müssen deshalb durch den Installateur je nach der Anwendungsart überarbeitet und realisiert werden.
El equipo esta diseñado para ser insertado como un componente de una máquina; las protecciones necesarias para la seguridad del operario deben de ser diseñadas y situadas por el instalador de acuerdo al tipo de trabajo en cuestión.
L'apparato è progettato per l'inserimento come componente di una macchina; pertanto, i dispositivi di sicurezza necessari per la protezione degli operatori devono essere progettati e realizzati dall'installatore, in funzione del tipo di applicazione.

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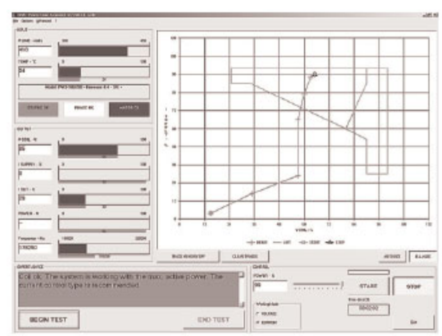
Resp. Laboratorio EMC / Resp. Lab. EMC

Doc. N° PW3-180/200 rev 4 | Mod. P0401-0025 rev 0

6.2 Spare parts

Parte di ricambio			Codice d'ordine
	Power Cube 90/50 generator		31282
	Power Cube 90/200 generator		31281
	Power Cube 180/50 generator	*12kW*	21255
		24kW *Configured for 2 operating modules in parallel*	54916
		48kW *Configured for 4 operating modules in parallel*	54917
	Power Cube 180/200 generator	*12kW*	30021
		24kW *Configured for 2 operating modules in parallel*	33044
		48kW *Configured for 4 operating modules in parallel*	41185
-	00815SCD Control Board		76516

6.3 PowerCube Assistant

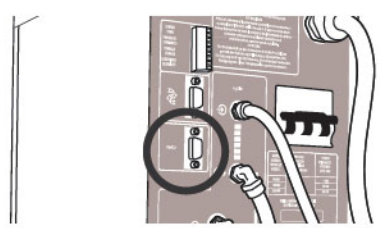


The PowerCube Assistant has been designed to optimize the heating coils, usable on series 50, 200 and 900 – 1800 generators.

The software is free and downloadable, upon registration, at the address:

http://www.ceia-power.com/secure/reserved.asp

The file is located in the Reserved Area of the site, accessible only after the registration.



To connect the PowerCube 90 and 180/XX generator to a computer running the PowerCube Assistant, follows the instructions below:

- Turn off the PowerCube 90 and 180 and disconnect it from all the supply sources.
- Connect the RS-232 port of your computer to the RS-232 port of the generator.

6.4 Control via RS232 serial interface

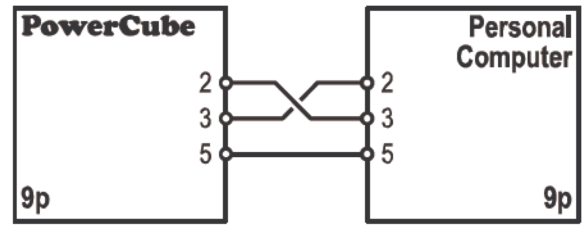
The generators of Power Cube family can be managed through the use of a serial line, using a Terminal or a Personal Computer.

6.4.1 Setting of the Terminal

Connect the RS232 port of the generator to a terminal configured with the following line parameters:

Baud rate	: 9600
parity	: no parity
data bit	: 8
stop bit	: 1
flow control	: none

For this connection a serial communication cable (CEIA code 19623 may be used).



6.4.2 Starting of the connection

Send the character CTRL-Z (ASCII 26) and wait for the character ">" to appear (should it not appear, repeat the operation).

6.4.3 Commands

The available commands are:

Syntax	Description
H1 <CR>	It activates the head 1 of the generator
S1 <CR>	It stops heating at head 1.
P1<CR>	Reading of the working power pre-selected for the heating head 1: a value from 10 to 99 is displayed which has to be intended as percentage of the maximum power.
P1=xx<CR>	The working power for the heating head 1 must be set at xx, being xx between 10 and 99. The pre-selected power can be varied only when the heating head is not activated.
PW<CR>	Reading of the power supplied: a value from 10 to 99 is displayed which has to be intended as percentage of the maximum power or -- if none of the two heating heads is activated.
SC<CR>	Reading of the POWERCUBE state: in the event that none anomalous situation is present **OK** is the answer displayed, otherwise a diagnosis code **Ax** will be displayed, having x between 1 and 12 showing an anomalous situation. See on section **MAINTENANCE** of the PowerCube instruction booklet for the codes meaning comprehension.

☞ In case of wrong syntax, out of range parameter or unrecognized command, the controller will show the message: ???

6.5 Manufacturer

CEIA S.p.A.
Costruzioni Elettroniche Industriali Automatismi
Zona Industriale 54/56, 52041 Viciomaggio – Arezzo (ITALY)

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